

## CHAPTER 8

# Disturbances

### 8.1. Space story

I was sitting at my desk when my boss came in and said, “You have to go to Japan now!” It turned out their satellite was losing roll control. I was able to delay a day but then had to fly from Newark to Tokyo and then take a taxi down to the Kimitsu ground station. The magnetic-torquer control system did not have enough control authority because it was undersized for December when the Sun suppressed the Earth’s magnetic field. The torquers operated in saturation. Fortunately, each torquer had a backup coil. When that was turned on there was just enough additional torque to control the spacecraft. The problem arose because of a faulty disturbance model that did not account for the Sun’s effect. The Sun causes the atmosphere to expand on the opposite side of the Earth. This distorts the magnetic field changing the field seen by the spacecraft.

### 8.2. Introduction

This chapter discusses torque and force disturbances on a spacecraft. Disturbances may be internal or external. External disturbances must be countered by actuators and are used to size actuators. The ultimate performance of the spacecraft will depend on the magnitude and direction of the disturbances. For those familiar with aircraft the magnitude of the disturbances, often on the order of micro-Newton meters ( $\mu\text{N m}$ ) seem very small. However, because they act on the spacecraft for a long period, they can cause considerable perturbations in the spacecraft orientation.

External disturbances are a function of the orientation of the spacecraft, its orbital position and velocity, the relative position of nearby planets, and the orientation of the Sun with respect to the spacecraft. Because external disturbances are due to interaction with the spacecraft the type of surfaces is also important. Surfaces can change once the spacecraft is on-orbit so these changes must also be considered. The torque is often a function of the center-of-mass of the spacecraft. Gravity-gradient torques are determined by the inertia matrix.

Internal disturbances can cause pointing errors and may also require either active or passive compensation. Internal disturbances typically come from moving parts in the spacecraft. Sometimes the control actuators, such as reaction wheels, are a source of disturbances. The reaction-wheel noise required a suspension on the Hubble Space Telescope and precluded their use entirely on the Gravity Probe B spacecraft.

Disturbances due to outside sources, such as gravity gradient, aerodynamic, and solar, are functions of the attitude or orientation and in the case of aerodynamic disturbances, the velocity of the spacecraft. They cause changes in the dynamical equations of the spacecraft and must be accounted for when modeling the dynamics. Fig. 8.1 shows how disturbance analysis fits into the design process. The column on the left lists the required inputs for modeling disturbances. The surface model is a surface mesh, usually consisting of triangles. There are two broad classes of disturbances, those that are dependent on the surface geometry and those that are independent of the surface geometry. All of the external-force models are a function of the orbit and other celestial objects such as the Sun or nearby planets or moons. Gravity gradient is dependent on the position of the satellite, the gravitational constant of the nearby planet, and the spacecraft inertia matrix. Residual dipole, radio-frequency disturbances, and outgassing depend on spacecraft properties.

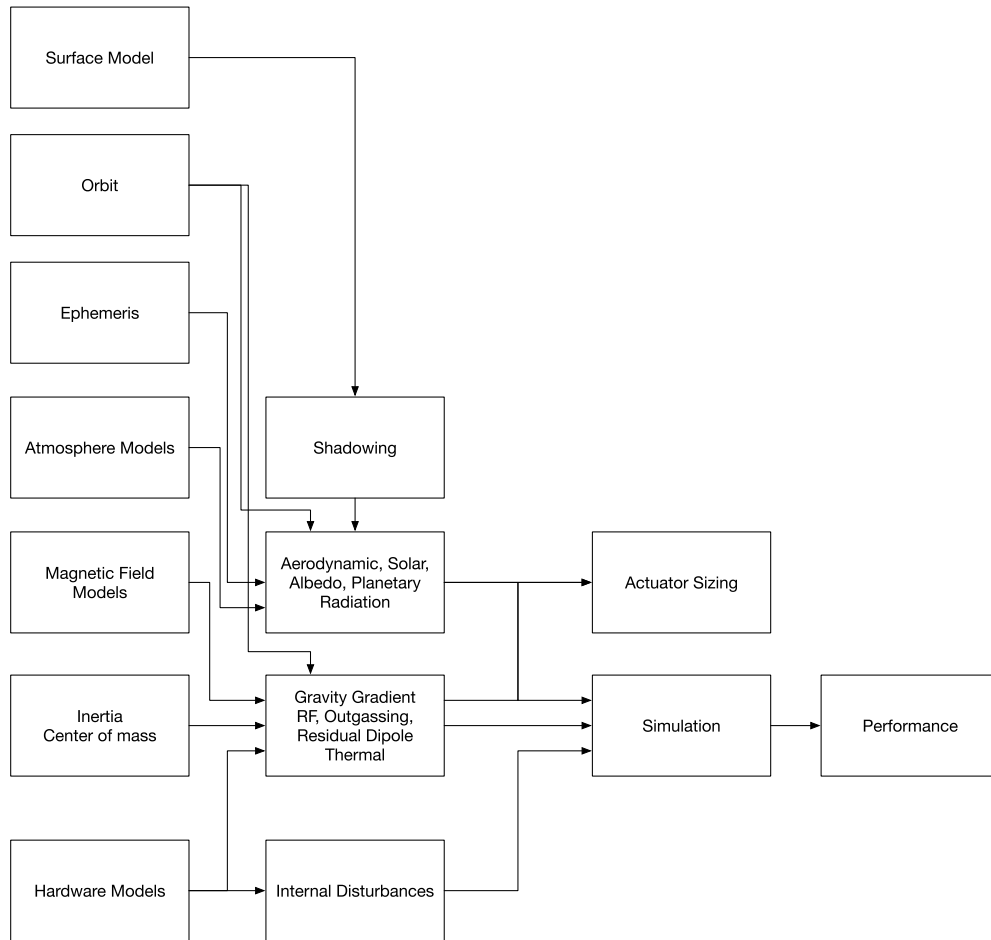
The blocks on the left are the information sources for disturbances. The ephemeris is the position of the Sun and other sources of fluxes. Eclipses are a function of the relative position of the Sun and nearby planets or asteroids. Atmosphere models determine the atmospheric density that drives the aerodynamics forces. The magnetic field determines the residual dipole torque and also the torque produced by any magnetic torquers. The inertia determines the gravity-gradient torque. The center-of-mass, combined with forces due to external fluxes, determines the torques due to the external fluxes. The spacecraft hardware determines internal torques and some forces and torques, such as outgassing and thermal.

For complex geometries, shadowing may have to be considered. Solar wings, antennas, and other deployed structures can shadow other parts of the spacecraft. This can have a large impact on the disturbances.

Disturbance modeling is used for two purposes. One is sizing actuators and the other is for performance simulations. The disturbances on the spacecraft are one factor in sizing momentum-exchange devices. The required torque must be larger than the instantaneous disturbance torque and the momentum storage is based on the steady inertial torque. If thrusters are used, the disturbance level will determine the required minimum impulse bit. Other factors, such as required maneuvers, also factor into actuator sizing.

Simulations do not require high-fidelity disturbance models. In some cases, if the spacecraft does not change its orientation, a Fourier-series representation can be used. If the spacecraft maneuvers, the disturbance model needs to be incorporated into the simulation. The model can be of lower fidelity than that used for the disturbance analysis.

This chapter will provide models for all external disturbances. Internal disturbances will be discussed in the context of the actuators in the chapter on actuators.



**Figure 8.1** Disturbance analysis fits into the spacecraft-design process in several places.

### 8.3. External disturbances

External disturbances change the angular momentum of the spacecraft. Disturbances can produce periodic or constant momentum changes. The former will cause the spacecraft to oscillate (or if controlled by reaction wheels) the wheel speed to oscillate at harmonics of the orbit rate. The latter will cause an increase in the spacecraft's angular momentum. This increase requires external torquers, typically provided by thrusters or magnetic torquers, to control the increase. It is necessary to tabulate all of the disturbances in a disturbance budget for the spacecraft. The budget, which has temporal categories, is used to size actuators and/or momentum storage. Models of the disturbances are also used in simulations of the spacecraft.

It may be surprising to talk about radiation pressure. In the early days of spaceflight, not everyone was convinced that light could produce a force. Table 8.1 lists major external disturbances on spacecraft. Some of these are discussed in the following sections.

**Table 8.1** Disturbances.

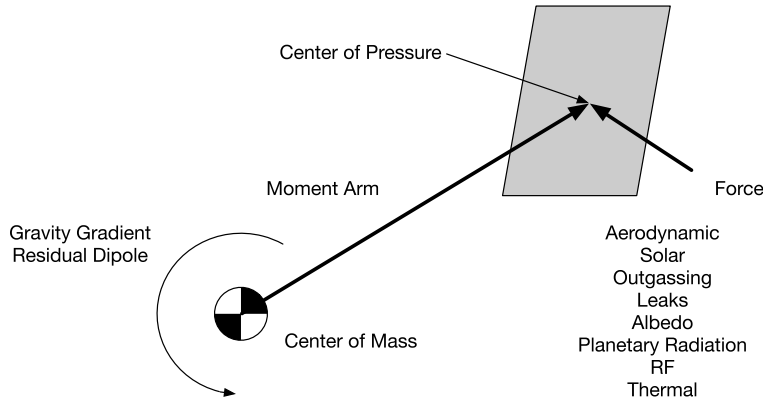
| Type                 | Source                                 | Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aerodynamic          | Planetary atmosphere.                  | Drag and lift on the spacecraft. A strong function of altitude and position relative to the Sun. Most spacecraft do not have lift. The surface of the spacecraft is very important.                                                                                                                                                                                                                                                                                                                |
| Albedo Pressure      | Sun reflection from the Earth.         | The average is 0.34 of the solar flux for the Earth. Albedo varies depending on the latitude and longitude of the spacecraft, season, and specific surface properties of the planet over which the spacecraft is flying. The relative angle of the Sun determines the direction of the force.                                                                                                                                                                                                      |
| Electrodynamic Force | Motion through Earth's magnetic field. | Produces drag or lift on a satellite. Varies depending on the strength of the magnetic field and the orientation of the satellite's path through the magnetic field. This is a very small effect and is rarely considered in a disturbance analysis.                                                                                                                                                                                                                                               |
| Gravity gradient     | Inertia and distance from the planet.  | This is due to the distributed nature of a satellite. If the satellite has off-diagonal inertia terms it will produce a body-fixed torque for a planet-pointing spacecraft. The diagonal terms cause a gravity-gradient modal frequency. A gravity-gradient-stabilized spacecraft uses a specific distribution of momentum to provide a stable oscillation. Gravity gradient is a pure stiffness term in the dynamical model. It cannot provide damping, which must be provided by another source. |
| Leaks                | Onboard gas and liquid supplies.       | Acts like a cold gas thruster. Leaks are not deliberate but may happen due to damage to the spacecraft. Pressurized spacecraft with human crews, must account for potential leaks. Waste dumps are a source of forces and torques. Spacecraft with crews are massive so these forces are not usually a major problem.                                                                                                                                                                              |
| Outgassing           | Moisture embedded in the structure.    | Caused by the heating of surfaces and the resulting emission of gas. This often happens with composite solar arrays and can produce large forces and torques. Outgassing is usually only an issue at the beginning of life for spacecraft.                                                                                                                                                                                                                                                         |

*continued on next page*

**Table 8.1** (continued)

| Type               | Source                                                  | Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Thruster Plumes    | Interaction of rocket exhaust plumes and the structure. | Thruster plumes are mathematically the same as aerodynamics forces and torques. They are modeled in two broad regimes, near-field and far-field. Far-field is when a thruster plume impinges on a surface far enough away from the thruster so that the flow is not a continuum. In the near-field, the flow is denser and fluid-dynamic models may be appropriate.                                                                                                                                      |
| Radiation Pressure | Temperature of the planet.                              | Black-body radiation of the planet. The value is 400 W for the Earth. This radiation pressure is at infrared wavelengths. When modeling planetary radiation, the optical properties at infrared wavelengths must be used.                                                                                                                                                                                                                                                                                |
| Residual Dipole    | Residual dipole on the spacecraft.                      | This torque is caused by the interaction of an external magnetic field and internal dipoles due to current loops, etc. Designers make every effort to eliminate current loops but they are not always avoidable. Magnetic torquers, using a torque rod, always have some residual dipole. Motors have magnetic materials and can also produce a residual dipole.                                                                                                                                         |
| RF                 | Transmit antennas.                                      | Radiation pressure. Divide the wattage by the speed of light to get the pressure. Communication satellites, with large offset transmitting antennas, can produce significant torquers on the spacecraft.                                                                                                                                                                                                                                                                                                 |
| Solar Pressure     | Solar radiation.                                        | The Sun produces large forces on the spacecraft. If the spacecraft is asymmetric, the offset of the center-of-mass and center-of-pressure will result in a large torque. This is why communication satellites usually have symmetric solar wings. The surface force depends on the surface properties. Some spacecraft, such as solar sails, take advantage of this force for propulsion. The solar flux is 1367 W at the Earth orbit and varies inversely with the square of the distance from the Sun. |
| Thermal Pressure   | Radiators.                                              | Heat radiates diffusely from all radiators. This produces a force proportional to the heat flux. The temperature of the spacecraft is not uniform. Some components, such as batteries, may have their own radiators.                                                                                                                                                                                                                                                                                     |

External disturbances either produce a force, which produces a torque due to a moment arm with respect to the center-of-mass, or produce a pure torque. This is shown in Fig. 8.2.



**Figure 8.2** Torque production. Some disturbances produce torques without a force.

### 8.3.1 Surface geometry

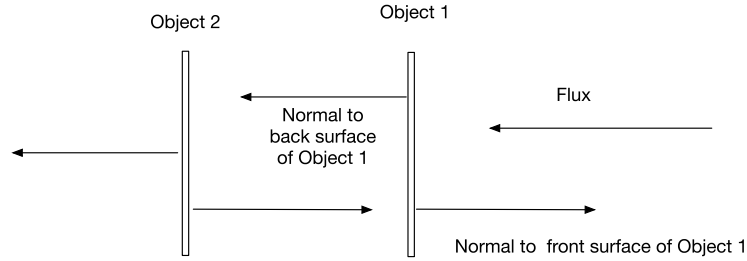
The surface mesh should be the simplest possible model that represents the surface. The resolution of the mesh should be such that the error due to mesh resolution is about the same order of magnitude as that of the other parameters in the model. Most spacecraft computer-aided design (CAD) packages generate hundreds of thousands of triangles and provide too much detail to be of use. An important consideration is to only include triangles that are visible to the incoming particle flux or radiation flux. Each surface element is associated with a normal that can be used to eliminate triangles that are not visible. However, care must be taken not to include internal surfaces. Fig. 8.3 shows two objects with front and back surfaces. One is blocked by the other. The normals for Object 1 can be used to select the faces of that object for disturbance calculation. The front normal of Object 2 cannot because it is shadowed.

### 8.3.2 Aerodynamic

#### 8.3.2.1 Simple drag coefficient

Aerodynamic disturbances are due to the interaction between a planetary atmosphere and a spacecraft's surface. It is assumed that the interaction is by a stream of particles that do not interact with each other. The simplest model for the aerodynamic force is

$$F = \frac{1}{2} \rho C_D A_p v |v| \quad (8.1)$$



**Figure 8.3** Normals can be used to select surfaces in some cases.

where  $\rho$  is the atmospheric density,  $C_D$  is the drag coefficient,  $A_p$  is the projected area, and  $\nu$  is the velocity vector. Lift is perpendicular to the velocity vector. This is the dynamic pressure multiplied by  $C_D A_p$ . The projected area for a flat plate is

$$A_p = A \cos \alpha \quad (8.2)$$

where  $\alpha$  is the angle between the surface normal and the velocity vector. Unlike an aircraft, only surfaces that interact with the particles that arrive with an angle between the velocity vector and the surface normal are considered.

This model assumes that the incoming particle is either reflected directly backward or absorbed by the surface. If it is reflected backward  $C_D$  is 4. If it is absorbed,  $C_D$  is 2.  $A$  is the projected area. If the flow is tangential to the surface, the drag is zero. Determining  $C_D$  is difficult. Estimates can be made from orbit-decay data, but this assumes the orientation is known and only measures  $\rho C_D$  at best.

It is useful to look at the magnitude of the forces. Assume the spacecraft has a surface area of  $1 \text{ m}^2$  and that the velocity vector is normal to the surface. Example 8.1 shows the drag on a satellite as a function of altitude for different atmosphere models. The exponential atmosphere, which is very popular because of its simplicity, produces a much lower drag force. Scale heights, essentially fit measurements of the atmosphere each over a restricted range, produce large and likely more realistic values. The example shows the problem that the atmosphere poses when modeling drag and computing aerodynamic disturbances.

### 8.3.2.2 Surface-accommodation coefficients

At altitudes above 120 km, the atmospheric density is sufficiently low that continuum flow does not apply. The Knudsen number shows the flow regime; less than 0.01 is continuum flow, greater than 10 is a free-molecular flow

$$Kn = \frac{\lambda}{L} \quad (8.3)$$

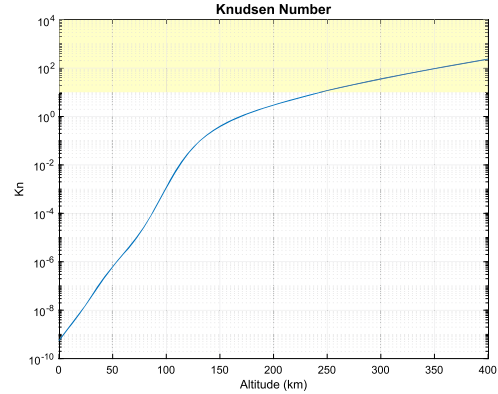




```

1 h = linspace(0,400);
2 rho = AtmDens2(h);
3 T = AtmTemp(h);
4 m = 28*1.66053886e-27;
5 kB = 1.38e-23;
6 mu = AbsoluteViscosity('n2', T);
7 l = 109;
8 kN = (mu./(rho*l)).*sqrt(pi*m./(2*kB*T));
9 Plot2D(h,kN,'Altitude_(km)', 'Kn', 'Knudsen_
    Number','ylog');
10 x = get(gca,'xlim');
11 y = get(gca,'ylim');
12 hold on
13 yL = 10;
14 fill([x(1) x(1) x(2) x(2)],[yL y(2) y(2) yL],[1 1
    0],'edgecolor','none','facealpha',0.2);

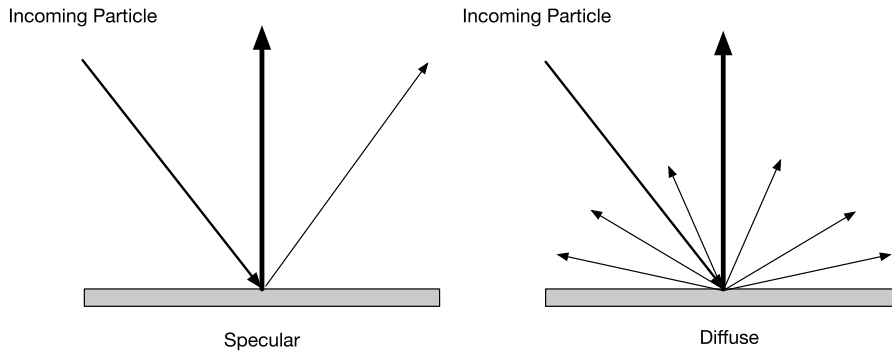
```



**Example 8.2:** Knudsen number. The shaded region is the free-molecular regime.

Free-molecular flow is shown in Fig. 8.4. An incoming particle strikes the surface. It can be absorbed, reflected specularly or reflected diffusely.

Surface-accommodation coefficients [1,2] incorporate these effects.



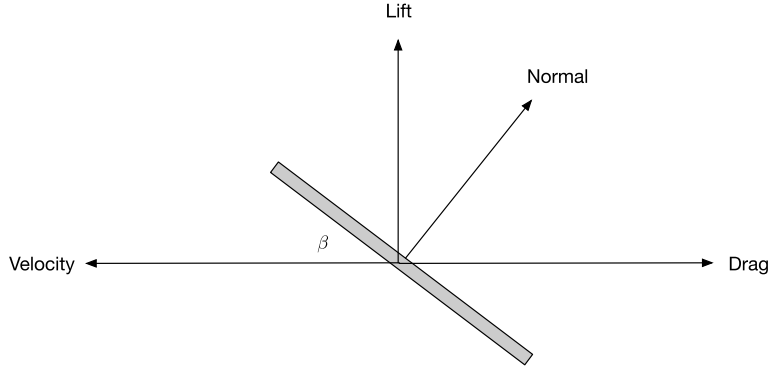
**Figure 8.4** Free molecular flow.

The surface-accommodation coefficients are

$$\sigma_n = \frac{p_i - p_r}{p_i - p_w} \quad (8.7)$$

$$\sigma_t = \frac{\tau_i - \tau_r}{\tau_i - \tau_w} \quad (8.8)$$

where  $p_i$  and  $p_r$  are the incident and reflected normal momentum flux,  $\tau_i$  and  $\tau_r$  are the incident and reflected tangential momentum flux,  $p_w$  and  $\tau_w$  are the normal and tangential components of momentum that would be carried away by diffusively emitted momentum, and  $\tau_w$  is by definition zero. For purely specular reflection with,  $p_i = p_r$  and  $\tau_i = \tau_r$ ,  $\sigma_n = \sigma_t = 0$ .



**Figure 8.5** Lift and drag on a plate.

There are two approaches to computing the aerodynamic forces. One assumes that the only contribution is the motion of the spacecraft. The second assumes that the particles in the atmosphere have a Maxwellian velocity distribution.

The average normal velocity of diffusely reflected molecules is

$$V_w = \sqrt{\frac{\pi R T_w}{2\mathfrak{M}}} \quad (8.9)$$

where  $R$  is the ideal gas constant, which is 8314.5 joules/kg-mole K,  $\mathfrak{M}$  is the mean molecular weight, and  $T$  is the gas temperature that can be found from the atmosphere model. The surface-accommodation coefficients are also a function of the incidence angle. Results obtained by bombarding an aluminum surface with nitrogen are [1]

$$\sigma_n = 0.93 - 1.48 \times 10^{-3}\theta - 7 \times 10^{-5}\theta^2 \quad (8.10)$$

$$\sigma_t = 0.63(1 - e^{-3.38 \times 10^{-2}\theta}) \quad (8.11)$$

Example 8.3 shows the normal and tangential accommodation coefficients.

Since the surface in the experiment was not molecularly clean, and all spacecraft surfaces are coated with a layer of monatomic oxygen, this result applies to most spacecraft surfaces.

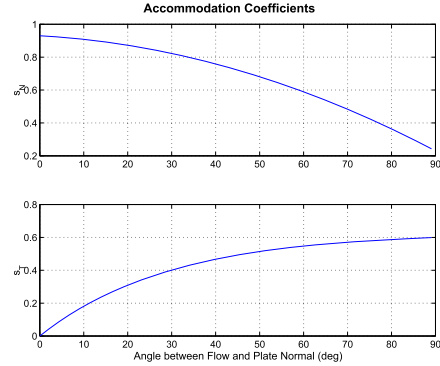
Fig. 8.5 shows the lift and drag geometry. For the zero atmospheric-motion case, we get

$$C_L = \left[ \sigma_n \frac{V_w}{V} + (2 - \sigma_n - \sigma_t) \sin \beta \right] \sin 2\beta \quad (8.12)$$

$$C_D = 2 \left[ \sigma_t + \sigma_n \frac{V_w}{V} \sin \beta + (2 - \sigma_n - \sigma_t) \sin^2 \beta \right] \sin \beta \quad (8.13)$$

For the specular case with  $\beta = \pi/2$ ,  $C_D = 4$ .

1 FAeroSurfaceAccommodation



### Example 8.3: Surface-accommodation coefficients.

With a Maxwellian velocity distribution, the equations are more complex

$$C_L = \left[ \frac{2 - \sigma_n - \sigma_t}{S\sqrt{\pi}} e^{-S^2 \sin^2 \beta} + \sigma_n \frac{V_w}{V} \right] \sin 2\beta \quad (8.14)$$

$$+ \left[ \frac{2 - \sigma_n}{S^2} + 2(2 - \sigma_n - \sigma_t) \sin^2 \beta \right] \cos \beta \operatorname{erf}(S \sin \beta) \quad (8.15)$$

$$C_D = 2\sigma_n \frac{V_w}{V} \sin^2 \beta + \frac{2}{\sqrt{\pi} S^2} \left[ (2 - \sigma_n) \sin^2 \beta + \sigma_t \cos^2 \beta \right] e^{-S^2 \sin^2 \beta} \quad (8.16)$$

$$+ 2 \left[ (2 - \sigma_n) \left( \sin^2 \beta + \frac{1}{2S^2} \right) + \sigma_t \cos^2 \beta \right] \sin \beta \operatorname{erf}(S \sin \beta) \quad (8.17)$$

where

$$S = \frac{V}{V_a} \quad (8.18)$$

and average velocity of the particles in the atmosphere is

$$V_a = \sqrt{\frac{\pi RT_a}{2\mathfrak{M}}} \quad (8.19)$$

where  $T_a$  is the atmosphere temperature.

Example 8.4 shows the two models. The results are similar.

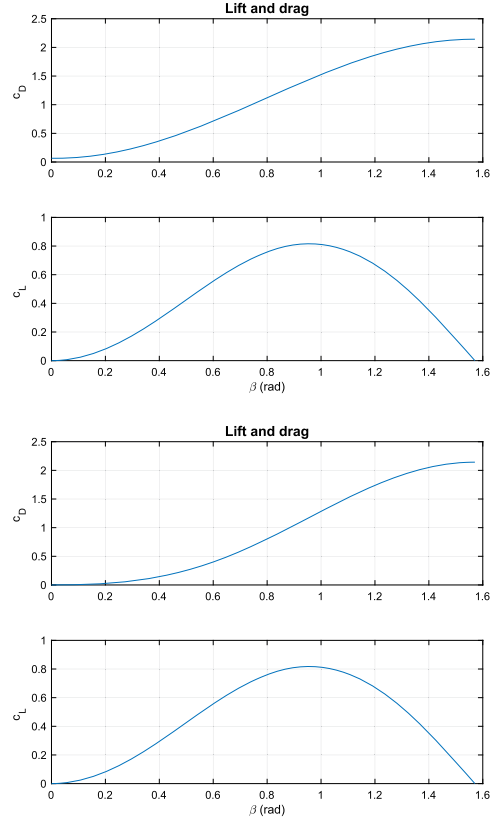
### 8.3.3 Electrodynamic force

An electrodynamic force is the result of the Lorentz force that occurs when a current-bearing wire moves through a magnetic field. In the context of satellite disturbances, the magnetic field is the geomagnetic field. This force can be exploited to produce either drag or lift for a satellite by an electrodynamic tether.

```

1 h = 350; % km
2 m = 28;
3 tA = AtmTemp(h);
4 tW = 300; % deg-K
5 v = sqrt(3.986e5/(6378+h))*1000;
6 beta = linspace(0,pi/2);
7 CLCDFreeMolecular( v, m, tW, beta )
8 CLCDFreeMolecular( v, m, tW, beta, tA )

```



**Example 8.4:** Free-molecular flow drag and lift coefficients. The bottom plot includes the Maxwellian atmospheric velocity.

Satellites attached to electrodynamic tethers experience electrodynamic forces either aligned with or opposing their motion perpendicular to the magnetic field. The force on the satellite is

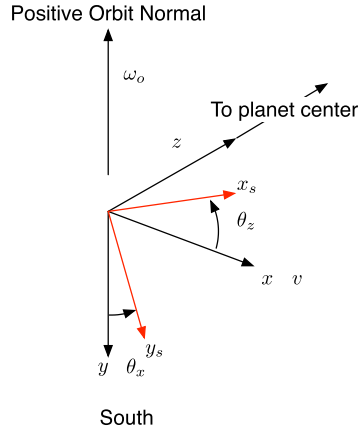
$$\mathbf{F} = I \int_0^l d\mathbf{l} \times \mathbf{B} \quad (8.20)$$

where  $I$  is the current in the wire and  $\mathbf{B}$  is the magnetic field. The current flowing through the tether will have a natural value that is a result of moving through the magnetic field that can be derived from Ohm's law,

$$I = \Phi / R \quad (8.21)$$

where  $\Phi$  is the voltage produced across the tether and  $R$  is the total resistance of the tether. For a satellite moving through the geomagnetic field with velocity  $\mathbf{v}$ , the voltage





**Figure 8.6** LVLH frame for the gravity-gradient analysis.

Assume only small rotations from the LVLH frame. Dividing through by the magnitude of  $r$  we get

$$T = 3\omega_o^2 u^\times I u \quad (8.25)$$

where the unit vector is

$$u = \frac{r}{|r|} \quad (8.26)$$

The  $r$  vector is along the LVLH  $z$ -axis. The skew-symmetric matrix is

$$u^\times = \begin{bmatrix} 0 & -u_z & u_y \\ u_z & 0 & -u_x \\ -u_y & u_x & 0 \end{bmatrix} \quad (8.27)$$

The unit vector in the body frame is

$$u = \begin{bmatrix} -\theta_y \\ \theta_x \\ -1 \end{bmatrix} \quad (8.28)$$

This is a “unit” vector if the angles are small. Expanding Eq. (8.25) we get

$$T = 3\omega_o^2 \begin{bmatrix} 0 & -1 & \theta_x \\ 1 & 0 & -\theta_y \\ -\theta_x & -\theta_y & 0 \end{bmatrix} \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{xy} & I_{yy} & I_{yz} \\ I_{xz} & I_{yz} & I_{zz} \end{bmatrix} \begin{bmatrix} -\theta_y \\ \theta_x \\ -1 \end{bmatrix} \quad (8.29)$$

Dropping nonlinear terms we get

$$T = 3\omega_o^2 \left( \begin{bmatrix} I_{zz} - I_{yy} & I_{xy} \\ I_{xy} & I_{zz} - I_{xx} \\ I_{xz} & I_{yz} \end{bmatrix} \begin{bmatrix} \theta_x \\ \theta_y \end{bmatrix} + \begin{bmatrix} I_{yz} \\ -I_{xz} \\ 0 \end{bmatrix} \right) \quad (8.30)$$

The gravity-gradient torque can provide stiffness, depending on the magnitudes of the diagonal terms. Note that these terms need to be combined with the Euler coupling terms to compute the complete gravity-gradient stiffness. The off-diagonal terms produce disturbance torques.

### 8.3.5 Residual dipole

The residual-dipole disturbances come from the interaction of the magnetic fields generated by current loops on the spacecraft and solar arrays with the Earth's magnetic field. The solar arrays and core spacecraft must be accounted for separately.

The simplest magnetic-field model is a tilted dipole model of the Earth's magnetic field. The second, suitable for orbits greater than 10 000 km, includes the effect of the Sun and is known as the Mead-Fairfield model [3]. Mead-Fairfield is a quantitative model, based on spacecraft magnetometer data, that perturbs a dipole model to account for the Sun activity. Input is the solar activity index. The third is a complete spherical harmonic expansion of the Earth's magnetic field. Example 8.6 shows the difference between the dipole and Meade-Fairfield model. The Mead-Fairfield model should be used for geosynchronous spacecraft.

The torque due to a residual dipole is

$$T = \left( \sum C_k M_k \right) \times B \quad (8.31)$$

where  $C_k$  transforms from the frame in which the dipole  $M_k$  is expressed to the core body frame and  $B$  is the Earth's magnetic field resolved into the body frame. Example 8.7 shows the case for a geosynchronous spacecraft with a Sun-pointing solar array and an Earth-pointing bus each with a 5 ATM<sup>2</sup> residual dipole along the  $x$ -axis.

Not all planets have magnetic fields or have very weak fields. Venus has a very weak field, for example. Jupiter, on the other hand, has a very large, intense, and complex field.

### 8.3.6 Radio-frequency forces

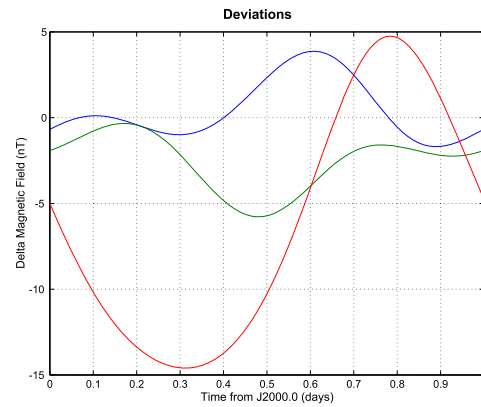
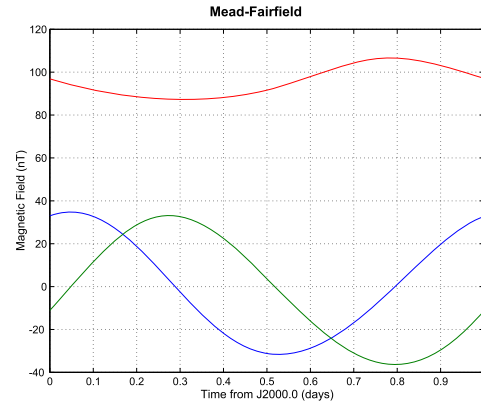
Transmitting antennas will produce torques. These are known as radio-frequency or RF torques. Any electromagnetic transmission will produce a force and a torque. The torque on the spacecraft will be

$$T = -\frac{p}{c} r^\times u \quad (8.32)$$

```

1 % Mead-Fairfield model of magnetic field
2 BMF

```

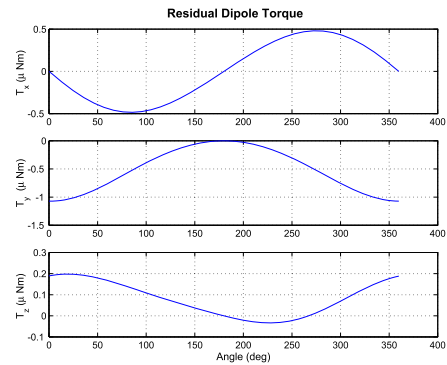


**Example 8.6:** Difference between a dipole and Meade–Fairfield magnetic-field model.

```

1 a = linspace(0,2*pi);
2 r = 42167*[cos(a);sin(a);zeros(1,100)];
3 jD = Date2JD( [2010 3 21 0 0 0] ) + linspace(0,1)
    ;
4 bT = BMF( r, jD );
5 n = length(a);
6 t = zeros(3,n);
7 d = [5;0;0]; % Dipole
8 for k = 1:n
9     c = cos(a(k));
10    s = sin(a(k));
11    m = [c s 0;-s c 0;0 0 1];
12    t(:,k) = Cross(d,m*bT(:,k)) + m*Cross(d,bT(:,k));
13 end
14
15 Plot2D( a*180/pi, t*1e6, 'Angle(deg)', ...
16         {'T_x(\mu Nm)', 'T_y(\mu Nm)', 'T_z(\mu Nm)'}, ...
17         'Residual_Dipole_Torque')

```



**Example 8.7:** Torque due to a residual dipole in geosynchronous orbit.



where  $p$  is the transmitted power,  $u$  is in the direction opposite the antenna boresight,  $r$  is the vector to the antenna boresight from the spacecraft center-of-mass, and  $c$  is the speed of light.

Internal RF transmissions, such as between the feeds and reflectors or in waveguides, do not produce a net torque on the spacecraft. Generally, RF poses a problem with high-power satellites with offset reflectors. For example, a transmitting antenna with an output of 600 W offset 3 m from the spacecraft center-of-mass will generate a 6- $\mu$ N m torque. Since this is a fixed torque over a week a spacecraft will accumulate 3.6 N m s from this disturbance alone.

### 8.3.7 Solar pressure

Solar pressure is the dominant disturbance on a spacecraft in geosynchronous and interplanetary orbit. Solar pressure is due to the force of photons on the surfaces of the spacecraft. The pressure is

$$s = \frac{p}{c} \quad (8.33)$$

where  $p$  is the power of the sunlight and  $c$  is the speed of light. As can be seen, a 1-N force would require  $3 \times 10^8$  W. Thus a flashlight propelled spacecraft would take a long time to gain any velocity.

A photon striking a surface can do one of four things: it can be absorbed, it can reflect specularly (meaning at the same angle with respect to the surface normal that it hit), it can reflect diffusely (meaning at any angle), or it can pass through the surface. Photons that are absorbed must either be transferred somewhere else (through heat conduction or as electricity in the case of a solar array) or be reemitted locally. If the latter happens, the photon must be lumped in with the diffusely reemitted photons. In terms of fractions of the incoming photons, the following is true for a surface

$$1 = \rho_a + \rho_s + \rho_d + \rho_t \quad (8.34)$$

where  $\rho$  stands for the fraction of photons that are absorbed, specularly reflected, diffusely reflected, or transmitted.

The solar pressure force is

$$F = -SA s^T n (2(\rho_s s^T n + \rho_d/3)n + (\rho_a + \rho_d)s) \quad (8.35)$$

where  $s$  is the Sun vector,  $n$  is the unit normal to the surface,  $A$  is the area of the surface, and  $S$  is the solar flux in N/m<sup>2</sup>. If the solar array tracks the Sun this becomes

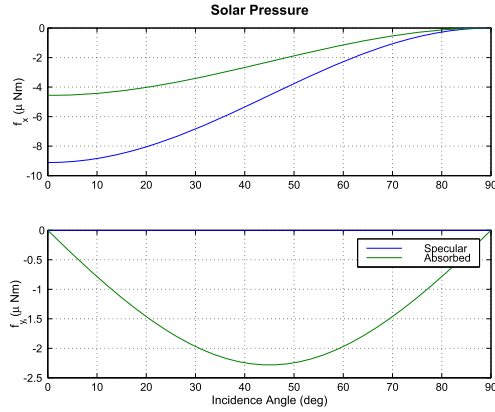
$$F = -SA(\rho_a + \frac{4}{3}\rho_d + 2\rho_s)s \quad (8.36)$$

The specular component produces the biggest force, they diffuse the second largest and the absorbed produces the smallest force contribution. A 1-m<sup>2</sup> surface produces the force shown in Example 8.8 in the cases of pure specular reflection and pure absorption.

```

1 a = linspace(0,pi/2);
2 n = length(a);
3 u = [cos(a); sin(a); zeros(1,n)];
4 p = 1367/3e8;
5
6 fS = zeros(3,n);
7 fA = zeros(3,n);
8
9 for k = 1:n
10 fS(:,k) = SolarF( p, [0;1;0;0], [1;0;0], u(:,k)
11 , 1)*1e6;
12 fA(:,k) = SolarF( p, [1;0;0;0], [1;0;0], u(:,k)
13 , 1)*1e6;
14 end
15 yL = {'f_x\(\mu Nm)'; 'f_y\(\mu Nm)'; 'f_z\(\mu Nm)'};
16 Plot2D( a*180/pi, [fS;fA], 'IncidenceAngle\(\deg)'
17 , yL, 'SolarPressure', ...,
18 'lin', {'[1;4]'; '[2;5]'} )
19 legend('Specular', 'Absorbed')

```



**Example 8.8:** Comparison between pure specular and pure absorption.

### 8.3.8 Earth-albedo force and torque

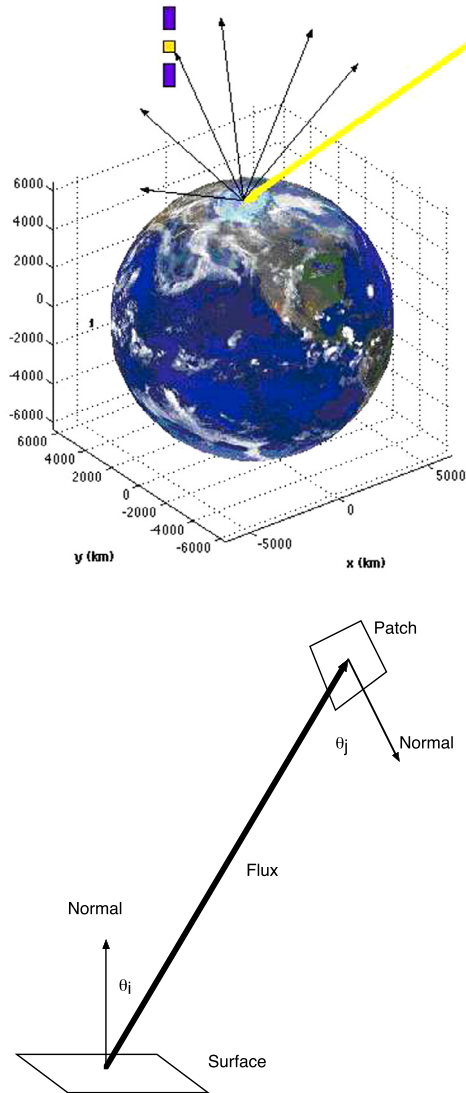
Albedo forces and torques are due to the reflection of sunlight off the surface and clouds of the Earth. Because the Earth cannot be treated as a point source (as can the Sun) and the reflectivity varies over the surface, albedo is difficult to model accurately [4]. The simplest model is to assume that reflection from the Earth's surface and clouds is diffuse [5]. The flux from the total reflectance can then be computed by integrating over the surface of the Earth.

The Earth's surface is modeled as a Lambertian reflector, as shown in Fig. 8.9.

The first step is to partition the Earth with a cloud map into triangles with each triangle being assigned a diffuse reflection coefficient. This is illustrated on the left of Fig. 8.7. A portion of the radiation reflected from the Earth or clouds intersects the surfaces of the spacecraft as shown on the right of Fig. 8.7. If the spacecraft is more than five times the distance from the surface as the dimensions of the surface, then the following equation can be used to model, the flux [6]

$$\Phi_{ij} = \alpha P A \frac{\cos \theta_i \cos \theta_j}{\pi r^2} \quad (8.37)$$

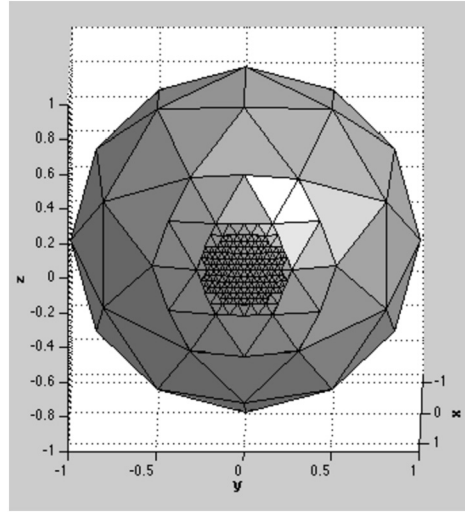
where  $\alpha$  is the absorption coefficient for the thermal flux,  $P$  is the solar flux,  $A$  is the area of the surface patch,  $\cos \theta_i$  is the angle between the normal of the surface patch and the incoming flux, and  $\cos \theta_j$  is the angle between the normal of the spacecraft patch



**Figure 8.7** Earth albedo and radiosity.

and the incoming radiation. For a spacecraft in a 200-km orbit, this would mean that the triangles representing the surface of the Earth could be no more than 40 km across.

This requires an active subdivision of the Earth's surface so that only the triangles under the spacecraft are tessellated. This is done using repeated tessellation of an icosahedron. An example is shown in Fig. 8.8. The  $\alpha$  coefficient varies with each patch and may represent ice, water, vegetation, clouds, etc. The spot under the spacecraft is sub-



**Figure 8.8** Adaptive subdivision.

divided into smaller triangles. The size of the triangles decreases with angular distance from the spacecraft vector.

### 8.3.9 Planetary-radiation force and torque

Planetary radiation is assumed uniform over the surface of the Earth. The flux at radius  $r$  is

$$q = \frac{W}{\left(\frac{r}{r_p}\right)^2} \quad (8.38)$$

where  $r_p$  is the radius of the planet and  $W$  is the radiation at the surface, which is 400 W for the Earth.

### 8.3.10 Thermal torque

The radiation emitted from a flat plate is

$$W = \sigma \epsilon A T^4 \quad (8.39)$$

where  $\sigma$  is the Boltzmann constant  $= 5.67 \times 10^{-12} \text{ W/cm}^2 \text{ K}^4$ ,  $\epsilon$  is the emissivity of the plate,  $A$  is the area, and  $T$  is the temperature in Kelvin. The radiation is emitted with a distribution proportional to  $\cos \phi$ , where  $\phi$  is the angle from the normal as illustrated in

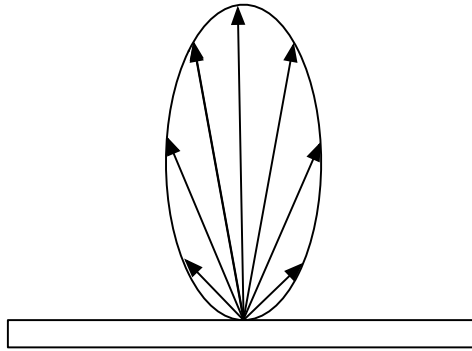
Fig. 8.7. Therefore the force coefficient will be

$$\frac{\int_0^{2\pi} \int_0^{\pi/2} \cos^2 \phi \sin \phi d\phi d\theta}{\int_0^{2\pi} \int_0^{\pi/2} \cos \phi \sin \phi d\phi d\theta} = \frac{\int_0^{\pi/2} \cos^2 \phi \sin \phi d\phi}{\int_0^{\pi/2} \cos \phi \sin \phi d\phi} = \frac{\left. \frac{\cos^3 \phi}{3} \right|_0^{\pi/2}}{\left. \frac{\cos^2 \phi}{4} \right|_0^{\pi/2}} = \frac{2}{3} \quad (8.40)$$

Therefore the force will be

$$F = \frac{2}{3} \frac{\sigma \epsilon A T^4}{c} \quad (8.41)$$

All surfaces will experience a force due to radiative emissions. In steady state, the surface temperature will be what balances radiative input with total output.



**Figure 8.9** Diffuse surface radiation.

If we define the thermal absorptivity of a surface as  $\alpha$ , then the steady-state thermal balance is

$$\alpha W = \sigma \epsilon A T^4 + q \quad (8.42)$$

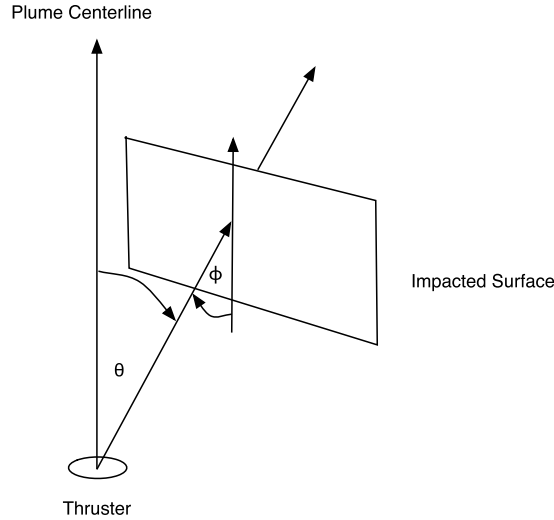
per unit area, where  $W$  is the incoming solar flux and  $q$  is the input or output from the rest of the spacecraft. The remainder of the flux,  $1 - \alpha$ , is either transmitted through the surface or reflected and it may be reflected either specularly or diffusely. The thermal force becomes

$$F = \frac{2}{3} \frac{A}{c} (\alpha W + q) \quad (8.43)$$

### 8.3.11 Thruster plumes

Thruster plumes cause disturbance torques and forces on the spacecraft when they impact spacecraft surfaces. Plume models are of two types, near-field and far-field. The former apply near the thruster nozzle when the exhaust gas can still be modeled as a

fluid. The latter model is applicable when the gas is best modeled as individual particles. The plume geometry is illustrated in Fig. 8.10.



**Figure 8.10** Plume-geometry diagram.

The plume is modeled as a far-field disturbance using the method of characteristics to model the plume-flow field. The plume disturbance is assumed to be due solely to interaction with the solar array. The solar array is broken up into panels and the torque is computed at the center of each panel. The plume torque is

$$T = Ar \times p \quad (8.44)$$

where  $A$  is the area of the panel,  $r$  is the vector from the spacecraft center-of-mass to the panel center-of-pressure, and  $p$  is the average plume pressure on the panel. The pressure vector is along the line from the thruster center to the panel center-of-pressure. The plume pressure has a component normal and a component tangential to the surface. The normal pressure component is along the panel antinormal (with the normal defined as pointing towards the thruster.) The tangential component is along the unit vector

$$\frac{n \times \frac{r \times n}{|r \times n|}}{|n|} \quad (8.45)$$

The total pressure, based on the density at the center of the panel, is

$$F = \frac{TA}{R^2 \Omega_{\text{eff}}} (\cos^4 \theta + 0.06) \cos^4 \theta \quad (8.46)$$

where  $T$  is the thruster thrust,  $R$  is the distance from the thruster to the center-of-pressure,  $\Omega_{\text{eff}}$  is the effective plume angle in radians (about  $\pi/2$ ), and  $\theta$  is as defined in the figure. The normal force is

$$F_N = P((2 - \sigma_N) \sin \phi + \sigma_N \beta) \sin \phi \quad (8.47)$$

and the tangential force is

$$F_T = P \sigma_T \cos \phi \sin \phi \quad (8.48)$$

where

$$\beta = \sqrt{\frac{\pi T_P}{T_0} \left( \frac{\gamma - 1}{4\gamma} \right)} \quad (8.49)$$

where  $\sigma_T$  and  $\sigma_N$  are the surface accommodation coefficients,  $T_P$  is the surface temperature of the panel,  $\gamma$  is the ratio of specific heats for the plume gas, and  $T_0$  is the combustion temperature of the plume. The accommodation coefficients are generally a function of  $\phi$  but can be set equal to 1 in the model to get the largest plume pressure.

### 8.3.12 Outgassing force

Outgassing is due to material loss due to diffusion from the spacecraft. The diffusion equation is [7]

$$\frac{dm}{dt} = -\frac{m}{\tau} \quad (8.50)$$

where

$$\tau = \tau_0 e^{\frac{E}{RT}} \quad (8.51)$$

where  $R$  is the universal gas constant for that molecule and  $T$  is temperature. The activation energy  $E$  is a function of the material but the reaction time constant  $\tau_0$  must be determined experimentally. Outgassed material may also impact a spacecraft surface and will stick, at least temporarily, to that surface. Outgassing has been responsible for spacecraft perturbations, including that on the Microwave Anisotropy Satellite [8].

The thrust due to outgassing is

$$T = \dot{m} u_e \quad (8.52)$$

where  $u_e$  is the velocity of the material. The exhaust velocity is [8]

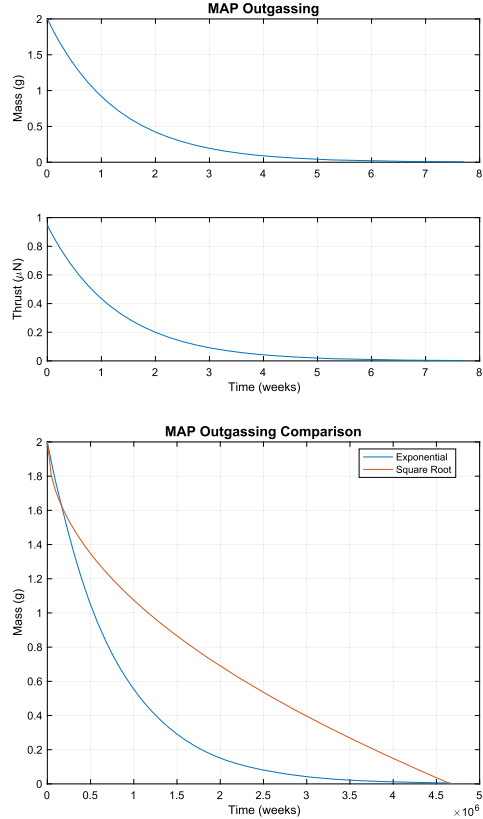
$$u_e = \sqrt{\frac{2k_b T}{N\mathfrak{M}}} \quad (8.53)$$

where  $\mathfrak{M}$  is the molecular mass,  $k_b = 1.38 \times 10^{-23}$  J/K,  $T$  is the temperature, and  $N$  is the atomic number of the outgassed material. Example 8.9 gives an example from the Microwave Anisotropy Spacecraft for water outgassing.

```

1 kB = 1.38e-23;
2 T = 150;
3 m = 1.7e-27;
4 N = 18;
5 uE = sqrt(2*kB*T/(m*N));
6 m0 = 0.002;
7 tau = 9*86400;
8 t = linspace(0,6*tau);
9 m = m0*exp(-t/tau);
10 mDot = m/tau;
11 thrust = mDot*uE*1e6;
12 R = 461.52;
13 EA = 26.4e3; % J/mole
14 tau0 = tau/exp(R*T/EA);
15 [tS,tL] = TimeLabl(t);
16 yL = {'Mass(g)', 'Thrust(\muN)'};
17 Plot2D(tS,[m*1000;thrust],tL,yL,'MAP_Outgassing')
18
19 g0 = 0.5*m0/(exp(EA/(R*T))*sqrt(6*tau));
20 mS = m0 - 2*g0*exp(EA/(R*T))*sqrt(t);
21 Plot2D(t,[m;mS]*1000,tL,yL{1},'MAP_Outgassing_
    Comparison','lin',...
22 [],[],[],[],{ {'Exponential' 'SquareRoot' }})

```



**Example 8.9:** Outgassing from the Microwave Anisotropy Spacecraft. Both exponential and square-root models are shown.

$\tau_0$  is 56 483 s. Soares [7] notes that diffusion has been experimentally shown to have a  $\sqrt{t}$  time dependence. The outgassing mass is

$$m = m_0 - 2q_0 e^{\frac{E}{RT}} \sqrt{t} \quad (8.54)$$

This, however, leads to an infinite diffusion rate at  $t = 0$  so is not useful for disturbance analyses.

### 8.3.13 Shadowing

An issue with external disturbances produced by external fluxes, whether aerodynamic or radiation, is that all surfaces will not be exposed to the flux because they are shadowed



by other surfaces. The problem of shadowing is encountered in computer graphics. It is solved by  $z$ -buffering, also known as depth buffering, which is simply drawing the objects from back to front along the view axis of the camera. For disturbance analyses, we need to be able to determine what part of a surface is visible behind a blocking surface. A simple, and reasonably fast algorithm, is the scan-line algorithm. This is also known as ray tracing. In this algorithm, the object is projected onto a plane normal to the flux vector and scanned. The objects are sorted, like in  $z$ -buffering, from front to back. If the scan line intercepts a front surface that surface is included. The scans that do not intercept the front surface are used to determine intersections with the next surface, and so forth. Fig. 8.11 shows an example with two triangles.

This method can be computationally intensive. The scan-line pattern must be chosen so that a scan line will pass through each triangle of a surface at least once. Complex geometries may cause scanning artifacts. The need for the computation of shadowing must be determined on a case-by-case basis. For really simple geometries, such as a CubeSat with a single deployable solar wing, it may be easier to analytically compute the shadowing.

## 8.4. Internal disturbances

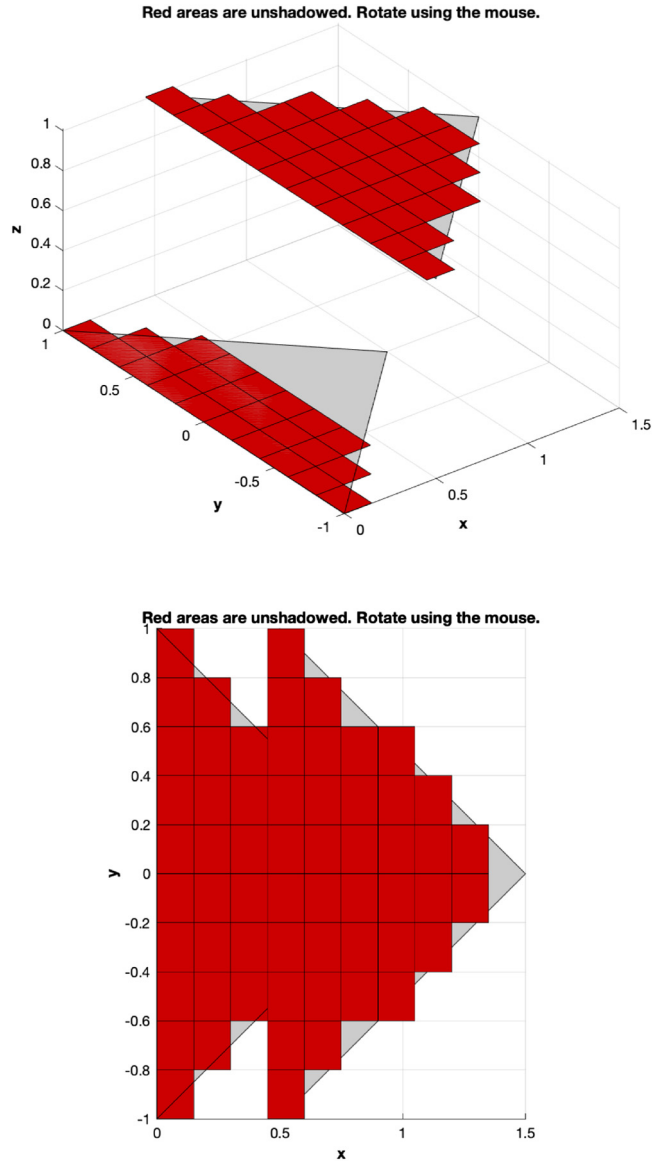
Table 8.3 lists major internal disturbances on spacecraft. Internal disturbances can cause jitter, which is high-frequency torques internal to the spacecraft.

The modeling of internal disturbances depends on the frequency of the disturbance. For example, with a reaction wheel, cogging torques, or imbalance torques are a function of the angular speed of the wheel. This varies as the wheel spins up or down. At low speeds, the torque might be within the bandwidth of the control system and would be modeled as a harmonic disturbance. At higher speeds, the torque would be modeled as noise. Internal dynamical disturbances can be modeled with multibody dynamics. For example, slosh motion during a burn can be modeled as a single or double pendulum. Friction torques for the reaction wheel are computed as part of the reaction-wheel model.

Thermal snap is due to the rapid change of temperature on a deployed part of the spacecraft, such as a solar wing or antenna causing it to vibrate. The Upper Atmosphere Research Satellite experienced thermal snap due to its single solar wing [9]. This effect can be modeled with multibody dynamics or as a single wheel with a harmonic torque.

## 8.5. Fourier-series representation

Fourier series can be used to represent the disturbances on a spacecraft. A complex numerical disturbance model is run and the output Fourier is analyzed to generate a harmonic series. The series can be used for simulations or actuator sizing. The Fourier



**Figure 8.11** Scan-line algorithm. The red rectangles are the included areas. The picture on the right shows the view down the flux vector.

series also gives insight into the nature of the disturbances.

$$T = a_o + \sum_k a_k \cos(knt) + b_k \sin(knt) \quad (8.55)$$

**Table 8.3** Internal disturbances.

| Type                     | Source                                              | Characteristics                                                                                                                                                                                                    |
|--------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dynamic imbalance        | Rotating component with off-diagonal inertia terms. | Periodic with the rotation rate.                                                                                                                                                                                   |
| Friction                 | Rotating or sliding parts.                          | Has three major components: stiction, or starting friction; Coulomb friction, which is independent of rate; and viscous, which is proportional to the rate. Turbulent damping is proportional to the rate squared. |
| Motor torques and forces | Motors.                                             | Articulating components.                                                                                                                                                                                           |
| Moving parts             | Dynamics of internal moving parts.                  | Torques are proportional to the square of relative rates.                                                                                                                                                          |
| Rotors                   | Momentum of rotors.                                 | Gyroscopic torques.                                                                                                                                                                                                |
| Rotor imbalances         | Center-of-mass offset of a rotating component.      | Periodic with the rotation rate.                                                                                                                                                                                   |
| Slosh                    | Liquids.                                            | Movement of liquids within the spacecraft. Important for liquid-fueled upper stages.                                                                                                                               |
| Thermal snap             | Sudden structural deformation.                      | Caused by rapid heating or cooling.                                                                                                                                                                                |

where  $n$  is the orbit period for a planet-pointing spacecraft. Any period associated with the spacecraft can be used. This is convenient for long-duration studies, particularly of momentum-control systems. It assumes that the spacecraft attitude motion is not coupled with the disturbances, which is true when a spacecraft is pointing tightly at a target.

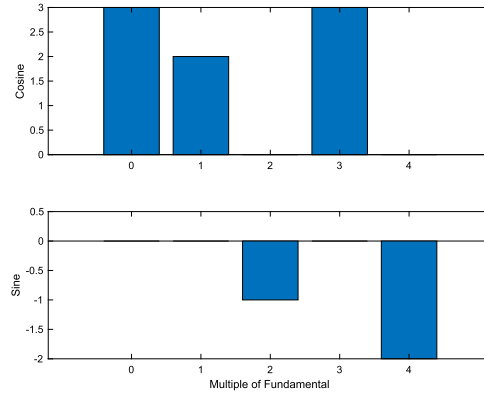
The discrete Fourier transform is a very efficient algorithm for generating the Fourier transform. Fast Fourier Transform (FFT) code is available in virtually every programming language. Example 8.10 shows how it is used. The code generates a vector of length  $n^2$ . The vectors are computed with sine and cosine terms with frequencies that are multiples of the fundamental frequency. The FFT is computed and half the values are discarded because they are duplicates of the first half. Each value corresponds to a harmonic that is a multiple 4 of the fundamental. The bias and cosine terms are from the real part of  $f$ . The sine terms are the negatives of the imaginary part. The coefficients are recovered exactly, even in the presence of noise.  $f$  is zero if there are no harmonics.

This can be applied to the problem of a geosynchronous satellite with a Sun-pointing solar wing and an Earth-pointing bus. This is shown in Example 8.11. The model has only five surfaces. Eclipses are included. A Fourier series is generated for the force. The spacecraft is modeled as a solar wing that is inertially fixed and a bus that is fixed with

```

1 %% Fourier Series
2
3 period = 10;
4 omega = 2*pi/period;
5 n = 2^12;
6 t = linspace(0,period,n+1);
7 y = 3 + 2*cos(omega*t) ...
8     - sin(2*omega*t) ...
9     + 3*cos(3*omega*t) ...
10    - 2*sin(4*omega*t) + randn(1,n+1);
11 f = fft(y,n);
12 f = f(1:n/2);
13 k = 0:(n/2-1);
14 a = 2*real(f)/n;
15 a(1) = a(1)/2;
16 b = -2*imag(f)/n;
17
18 figure('name','Fourier_Coefficients')
19 subplot(2,1,1)
20 bar(k(1:5),a(1:5))
21 ylabel('Cosine')
22
23 subplot(2,1,2)
24 bar(k(1:5),b(1:5))
25 xlabel('Multiple_of_Fundamental')
26 ylabel('Sine')

```



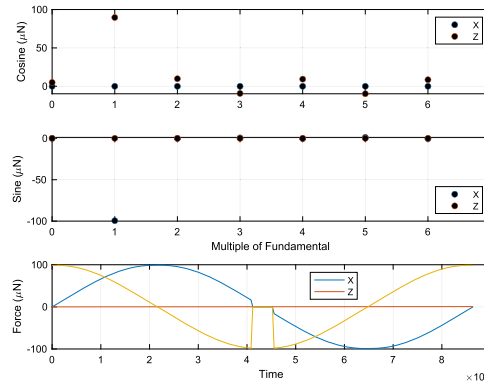
**Example 8.10:** Fourier coefficients of a harmonic function.

respect to LVLH. Spring equinox is chosen as the time of year so that the Sun is in the LVLH  $xz$ -plane. The spacecraft is in geosynchronous orbit and eclipses are modeled.

```

1 period = 86400; % s
2 areaWing = 2*10; % m^2
3 areaBus = 2*2; % m^2
4 rhoW = [0.5 0.1 0.3];
5 rhoB = [0.5 0.4 0.1];
6 u = [1 -1 0 0; 0 0 0 0; 0 0 1 -1];
7 p = 1367/3e8;
8 n = 2^8;
9 t = linspace(0,period,n+1);
10 jD = 2464042.0515 + t/86400;
11 [uSun,rSun] = SunV1(jD);
12 [r,v] = RVOrgen([42167 0 0 0 0],t);
13 e = Eclipse(r, uSun.*rSun);
14 q = QVLH(r,v);
15 s = uSun.*e;
16 %% Compute the force in the ECI frame
17 fW = -p*areaWing*(rhoW(3) + (4/3)*rhoW
18     (2) + rhoW(1))*s;
19 sB = QForm(q,s);
20 fB = zeros(3,n+1);
21 %% Cycle through the faces
22 for k = 1:4
23     uK = u(:,k);
24     uS = uK'*sB;
25     fB = fB - p*areaBus*uS.*(rhoW(3)*uS + rhoW
26         (2)/3).*uK + (rhoW(2)+rhoW(3))*sB);
27 end
28 %% Fourier analysis
29 fT = (fB + QForm(q,fW))*1e3;
30 f = fft(fT',n)';
31 f = f([1 3],1:n/2);
32 k = 0:(n/2-1);
33 a = 2*real(f)/n;
34 a(:,1) = a(:,1)/2;
35 b = -2*imag(f)/n;

```



**Example 8.11:** Fourier coefficients for a spacecraft in geosynchronous orbit.

The Fourier code is the same as the code in the example above. The force is dominated by the wing, which produces a sinusoidal force, even with the discontinuity due to the eclipse. The eclipse is evident in the middle of the plot when the Sun is on the opposite side of the Earth from the spacecraft. The force is dominated by the first harmonic, which is the orbit-rate term. The  $Y$  forces, which are zero because the Sun is in the  $xz$ -plane, are not plotted.

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